Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Densities of two gases are in the ratio $1:2$ and their temperatures are in the ratio $2:1$ then the ratio of their respective pressures is

(A) $1:1$

(B) $1:2$

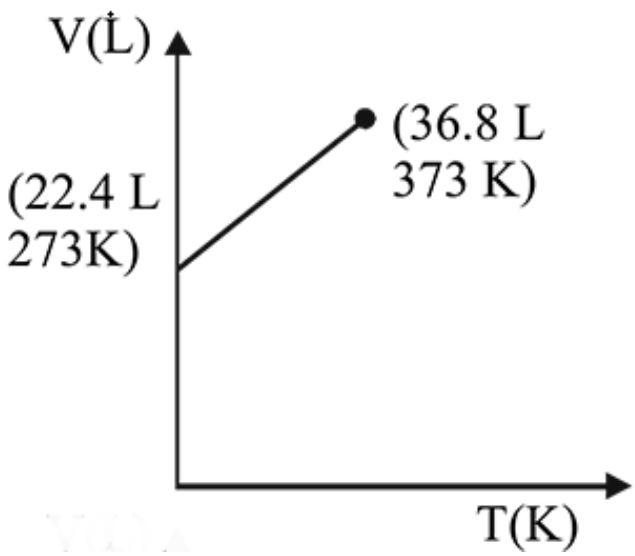
(C) $2:1$

(D) $4:1$

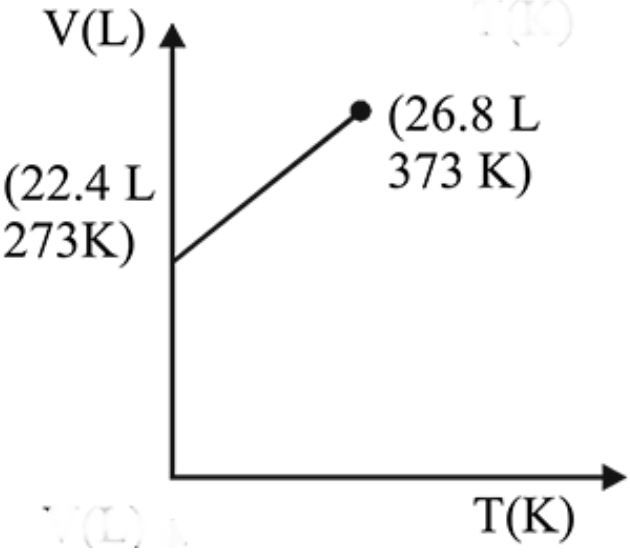
Q.2

Which of the following volume (V) - temperature (T) plots represents the behaviour of one mole of an ideal gas at one atmospheric pressure ?

(A)

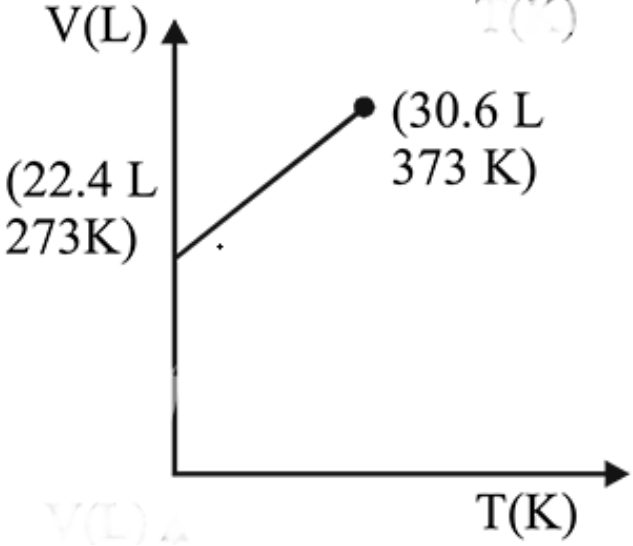


(B)

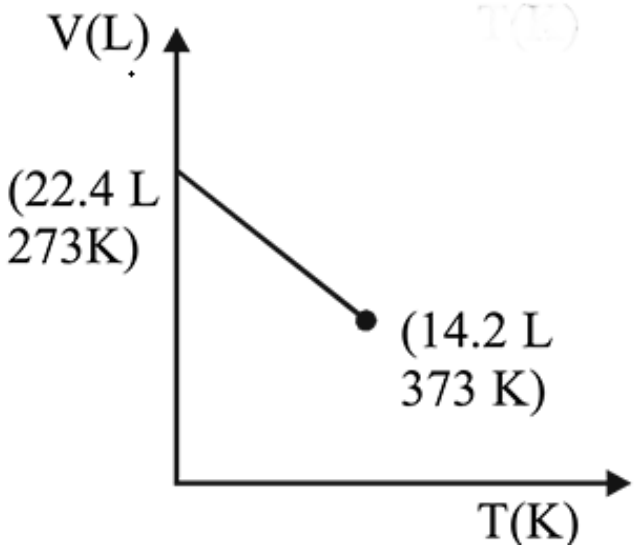


@IITJEE_Advanced

(C)



(D)



Q.3

600 c.c. of a gas at a pressure of $750\ mm$ of Hg is compressed to 500 c.c. Taking the temperature to remain constant, the increase in pressure, is

(A) $150\ mm$ of Hg

(B) $250\ mm$ of Hg

(C) $350\ mm$ of Hg

(D) $450\ mm$ of Hg

Q.4

The volume of $0.0168\ mol$ of O_2 obtained by decomposition of $KClO_3$ and collected by displacement of water is $428\ mL$ at a pressure of $754\ mmHg$ at $25^\circ C$. The pressure of water vapour at $25^\circ C$ is

(A) $18\ mmHg$

(B) $20\ mmHg$

(C) $22\ mmHg$

(D) $24\ mmHg$.

@IITJEE_Advanced

Q.5

The initial volume of a gas cylinder is $750.0\ \text{mL}$. If the pressure of gas inside the cylinder changes from $840.0\ \text{mmHg}$ to $360.0\ \text{mmHg}$, the final volume the gas will be:

(A) $1.750\ \text{L}$

(B) $3.60\ \text{L}$

(C) $4.032\ \text{L}$

(D) $7.50\ \text{L}$

Q.6

A bubble of the gas released at the bottom of a lake increases to eight times the original volume when it reaches at the surface. Assuming that the atmospheric pressure is equivalent to pressure exerted by a column of water $10\ \text{m}$ high, what is the depth of the lake

(A) $80\ \text{m}$

(B) $90\ \text{m}$

(C) $10\ \text{m}$

(D) $70\ \text{m}$

Q.7

A spherical balloon of $21\ \text{cm}$ diameter is to be filled with H_2 at NTP from a cylinder containing the gas at $20\ \text{atm}$ at 27°C. The cylinder can hold 2.82 L of water at NTP. The number of balloons that can be filled up is

(A) 15

(B) 10

(C) 20

(D) 25

Q.8

Equal volumes of the gases which do not react together are confined in separate vessels. The pressure is $200\ \text{mm}$ and $400\ \text{mm}$ of Hg respectively. If the two gases are mixed together what will be the pressure of the resulting mixture (temperature remaining constant)

(A) $400\ \text{mm}$

(B) $\sqrt{400}\ \text{mm}$

(C) $300\ \text{mm}$

(D) $200\ \text{mm}$

@IITJEE_Advanced

Q.9

When a sample of gas is compressed at constant temperature from $15\ \text{atm}$ to $60\ \text{atm}$, its volume changes from $76\ \text{cm}^3$ to $20.5\ \text{cm}^3$. Which of the following statements are possible explanations of this behaviour?
(1) The gas behaves non-ideally
(2) The gas dimerises
(3) The gas is adsorbed into the vessel walls

(A) 1,2 and 3

(B) 1 and 2 only

(C) 2 and 3 only

(D) 1 only

Q.10

At 27°C a sample of ammonia gas exerts a pressure of $5.3\ \text{atm}$. What is the pressure when the volume of the gas is reduced to one-tenth of the original value at the same temperature?

(A) $0.53\ \text{atm}$

(B) $5.3\ \text{atm}$

(C) $53\ \text{atm}$

(D) None of these

Q.11

A compressed cylinder of gas contains $1.50 \times 10^3\ \text{g}$ of N_2 gas at a pressure of $2.0 \times 10^7\ \text{Pa}$ and a temperature of 17.1°C. What volume of gas has been released into the atmosphere if the final pressure in the cylinder is $1.80 \times 10^5\ \text{Pa}$? Assume ideal behaviour and that the gas temperature is unchanged.

(A) $1264\ \text{L}$

(B) $126\ \text{L}$

(C) $12600\ \text{L}$

(D) $45\ \text{L}$

Q.12

A graph is plotted between PV_m (along Y-axis) and P (along X -axis) at 27° (V_m being the molar volume of an ideal gas). The intercept of the graph at Y -axis is $\left(\text{R} = 0.082\ \text{atmLK}^{-1}\ \text{mol}^{-1}\right)$

(A) $22.4\ \text{atm L mol}^{-1}$

(B) $24.6\ \text{atmLmol}^{-1}$

(C) $24.6\ \text{atmLK}^{-1}\ \text{mol}^{-1}$

@IITJEE_Advanced

(D) None of these

Q.13

Reducing the pressure from $1.0\ \text{atm}$ to $0.5\ \text{atm}$ would change the number of molecules in one mole of ammonia to

(A) 25% of its initial value

(B) 50% of its initial value

(C) 75% of its initial value

(D) None of the above

Q.14

Cyclopropane and oxygen at partial pressures 170 torr and 570 torr respectively are mixed in a gas cylinder. What is the ratio of the number of moles of cyclopropane to the number of moles of oxygen $(n\text{C}_3\text{H}_6/n\text{O}_2)$?

(A) $\frac{170\times42}{570\times32} = 0.39$

(B) $\frac{170}{42}/\left(\frac{170}{42}+\frac{570}{32}\right) \approx 0.19$

(C) $\frac{170}{740} = 0.23$

(D) $\frac{170}{570} = 0.30$

Q.15

A sample of gas occupies 100 litres at 1 atm pressure and at 0°C. If the volume of the gas is to be reduced to 5 litres at the same temperature, what additional pressure must be applied?

(A) $18\ \text{atm}$

(B) $19\ \text{atm}$

(C) 21 atm

(D) 10 atm

Q.16

A balloon of diameter $20\ \text{m}$ weights $100\ \text{kg}$. What is its pay - load, if it is filled with He at 1 atm and 27°C. Density of air is $1.2\ \text{kg m}^{-3}$

(A) $424.67 \times 10^4\ \text{g}$

(B) $424.67 \times 10^2\ \text{g}$

(C) $424.67 \times 10^5\ \text{g}$

(D) None of these

@IITJEE_Advanced

Q.17

In what ratio by mass carbon monoxide and nitrogen should be mixed se that partal pressure exerted by each is same?

(A) $1:1$

(B) $1:2$

(C) 2:1

(D) $3:4$

Q.18

Equal masses of $\mathrm{H}_2, \mathrm{O}_2$ and methane have been taken in a container of volume V at temperature $27^{\circ}\mathrm{C}$ in identical conditions. The ratio of the volumes of gases $\mathrm{H}_2 : \mathrm{O}_2$:
methane would be :

(A) $8:16:1$

(B) $16:8:1$

(C) $16:1:2$

(D) $8:1:2$

Q.19

For $1\ \mathrm{mol}$ of an ideal gas at a constant temperature T, the
plot of $(\log P)$ against $(\log V)$ is a (P : Pressure, V : Volume)

(A) Straight line parallel to x -axis.

(B) Straight line with a negative slope.

(C) Curve starting at origin.

(D) Straight line passing through origin.

Q.20

When $\mathrm{CO}_2(\mathrm{~g})$ is passed over red hot coke it partially gets reduced to $\mathrm{CO}(g)$. Upon passing $0.5\ \mathrm{L}$ of $\mathrm{CO}_2(g)$ over red
hot coke, the total volume of the gases increased to $700\ \mathrm{mL}$.
The composition of the gaseous mixture at STP is

(A) $\mathrm{CO}_2 = 300\ \mathrm{mL}; \mathrm{CO} = 400\ \mathrm{mL}$

(B) $\mathrm{CO}_2 = 0.0\ \mathrm{mL}; \mathrm{CO} = 700\ \mathrm{mL}$

(C) $\mathrm{CO}_2 = 200\ \mathrm{mL}; \mathrm{CO} = 500\ \mathrm{mL}$

(D) $\mathrm{CO}_2 = 350\ \mathrm{mL}; \mathrm{CO} = 350\ \mathrm{mL}$

@IITJEE_Advanced

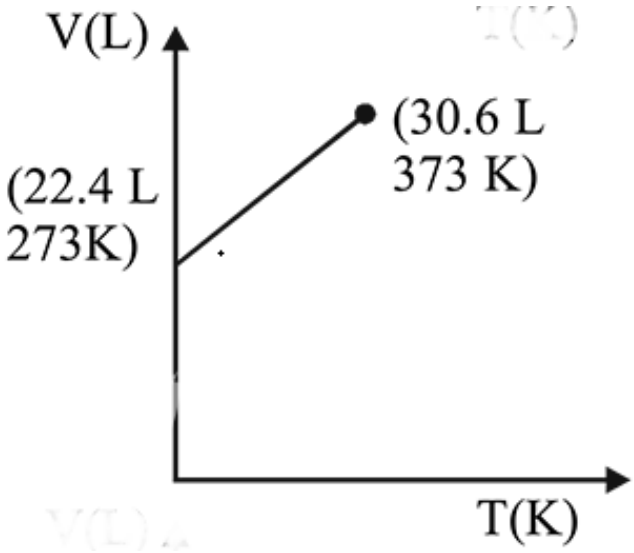
Q.1 Answer:

$1 : 1$

Solution:

$P \propto d$ **and** T,
$\frac{P_1}{P_2} = \frac{d_1\, T_1}{d_2\, T_2} = \frac{1}{2} \times \frac{2}{1} \Rightarrow P_1 : P_2 = 1 : 1$

Q.2 Answer:



Solution:

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ **at const. pressure**
$\Rightarrow \frac{22.4}{273} = \frac{V_2}{373}$, $V_2 = 30.6$ **litre**

Q.3 Answer:

150 mm of Hg

Solution:

Given initial volume $(V_1) = 600$ **c.c.; Initial pressure** $(P_1) = 750$ mm **of** Hg **and final volume** $(V_2) = 500$ **c.c. according to Boyle's law,** $P_1\, V_1 = P_2\, V_2$
or $750 \times 600 = P_2 \times 500$
or $P_2 = \frac{750 \times 600}{500} = 900$ mm **of** Hg
Therefore increase of pressure $= (900 - 750) = 150$ mm **of** Hg

Q.4 Answer:

24 mmHg.

Solution:

Volume of 0.0168 **moles at STP** $= 0.0168 \times 22400 = 376.3$ mL.
$\frac{P_1\, V_1}{T_1} = \frac{P_2\, V_2}{T_2}$
or $\frac{760 \times 376.3}{273} = \frac{P_2 \times 428}{298}$

@IITJEE_Advanced

or $P_2 = 730$ mm

Pressure of water $= 754 - 730 = 24$ mmHg

Q.5 Answer:

1.750 L

Solution:

According to Boyle's law

$$\frac{V_1}{V_2} = \frac{P_2}{P_1}; \frac{750}{V_2} = \frac{360}{840}$$
$$V_2 = 1750 \text{ mL} = 1.750 \text{ L}$$

Q.6 Answer:

70 m

Solution:

Let V **be the original volume of bubble. The final volume will be** 8 V**. Let** p **be the atmospheric pressure and** p_1 **the pressure at the bottom**

$\therefore \quad p \times 8\ V = p_1 \times V$

$$8p = p_1$$

p_1 **= atmospheric pressure +** p **pressure due to water lake** $= p + 7p$

The $p = 10$ m **high, the** $7p$ **will be** $= 70$ m **high So the depth** $= 70$ m

Q.7 Answer:

10

Solution:

The volume of the balloon $= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{21}{2}\right)^3 = 4851$ mL

Volume of the cylinder $= 2820$ mL

Volume of H_2 **at NTP**

$= \frac{20 \times 2820 \times 273}{300 \times 1}$ mL $= 51324$ mL

After filling the cylinder will have H_2 **equal to its volume** $= 2820$ mL

$\therefore$ **Volume of** H_2 **for filling balloons** $= 51324 - 2820 = 48504$ mL

Hence no. of balloon to be filled

$= \frac{48504}{4851} \approx 10$

Q.8 Answer:

300 mm

Solution:

@IITJEE_Advanced

When vessels are joined the volume is doubled and pressure is reduced to half Pressure of mixture

$$= \frac{200}{2} + \frac{400}{2} = 300 \text{ mm}$$

Q.9 Answer:

1 only

Solution:

Given, $P_1 = 15 \text{ atm}, P_2 = 60 \text{ atm}$

$V_1 = 76 \text{ cm}^3,\ V_2 = 20.5 \text{ cm}^3$

If the gas is an ideal gas, then according to Boyle's law, it must follow the equation, $P_1\ V_1 = P_2\ V_2$

$P_1 \times V_1 = 15 \times 76 = 1140$

$P_2 \times V_2 = 60 \times 20.5 = 1230$

$\therefore P_1\ V_1 \neq P_2\ V_2$

$\therefore$ The gas behaves non-ideally. The given information is not sufficient to comment on other statements.

Q.10 Answer:

53 atm

Solution:

$P_1\ V_1 = P_2\ V_2$

$V_1 = V, V_2 = V/10$

$\frac{5.3 \text{ atm}}{P_2} = \frac{\frac{V}{10}}{V}; P_2 = 5.3 \times 10 = 53 \text{ atm}$

Q.11 Answer:

1264 L

Solution:

$n_{N_2} = 53.57$ **mole**

$\frac{P_1}{P_2} = \frac{n_1}{n_2}$

$n_2 = \frac{1.8}{200} = 0.009 \times 53.57 = 0.48213$

n_{N_2} **escape** $= 53.087$

Volume $= \frac{nRT}{P} = 1264 \text{ L}$

Q.12 Answer:

$24.6 \text{ atmLmol}^{-1}$

Solution:

The graph is a straight line parallel to the axis of pressure with intercept $= PV_m = RT = 0.082$ $\text{atmLK}^{-1} \text{ mol}^{-1} \times 300 \text{ K}$

@IITJEE_Advanced

$= 24.6\ \text{atmLmol}^{-1}$.

Q.13 Answer:

None of the above

Solution:

One mole of a substance contains the number of molecules which is independent of pressure.

Q.14 Answer:

$\frac{170}{570} = 0.30$

Solution:

By ideal gas equation,

$P_1V = n_1RT$

$n_1 \propto P_1$ **and** $n_2 \propto P_2$

$\frac{n_1}{n_2} = \frac{P_1}{P_2} \Rightarrow \frac{n_1}{n_2} = \frac{170}{570} = 0.30$

Q.15 Answer:

19 atm

Solution:

Here $\mathrm{P}_1 = 1\ \text{atm} \quad \mathrm{P}_2 = ?$

$\mathrm{V}_1 = 100$ **litre** $\quad \mathrm{V}_2 = 5$ **litre**

$\mathrm{T}_1 = 273\ \mathrm{K} \quad \mathrm{T}_2 = 273\ \mathrm{K}$

As temperature is constant

Then from Boyle's law

$\mathbf{P}_1\mathbf{V}_1 = \mathbf{P}_2\mathbf{V}_2$

or $1 \times 100 = \mathbf{P}_2 \times 5$

$\therefore \mathrm{P}_2 = 20\ \text{atm}$

$\therefore$ **Additional pressure that should be applied** $= \mathbf{P}_2 - \mathbf{P}_1 = 20 - 1 = 19\ \text{atm}$

Q.16 Answer:

424.67×10^4 g

Solution:

Weight of balloon $= 100\ \text{kg} = 10 \times 10^4\ \text{g}$

Volume of balloon $= \frac{4}{3}\pi^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{20}{2} \times 100\right)^3 = 4190 \times 10^3\ \text{L}$

Weight of gas (He) in balloon

$= \frac{PV_m}{RT} = \frac{1 \times 4190 \times 10^3 \times 4}{0.082 \times 300} = 68.13 \times 10^4\ \text{g}$

$\therefore$ **Total weight of gas and balloon** $= 68.13 \times 10^4 + 10 \times 10^4 = 78.13 \times 10^4\ \text{g}$

Weight of air displaced $= \frac{1.2 \times 4190 \times 10^6}{10^3} = 502.8 \times 10^4\ \text{g}$

@IITJEE_Advanced

Payload = weight of air displaced - (weight of balloon + weight of gas)

Pay load $= 502.8 \times 10^4 - 78.13 \times 10^4 = 424.67 \times 10^4$ g

Q.17 Answer:

$1 : 1$

Solution:

$\mathrm{CO} = \mathbf{w}_1$ g $= \frac{\mathbf{w}_1}{28}$ mol, $\mathrm{N}_2 = \mathbf{w}_2$ g $= \frac{\mathrm{w}_2}{28}$ mol

$P_{\mathrm{CO}} = \frac{\frac{\mathbf{w}_2}{28}}{\frac{w_1}{28}+\frac{w_2}{28}} = \frac{w_1}{w_1+w_2}$

$\therefore \mathbf{P}_{\mathrm{N}_2} = \frac{\mathbf{w}_2}{\mathbf{w}_1+\mathbf{w}_2}$ **as** $\mathbf{P} = \mathbf{P}_{\mathrm{N}_2}$

Hence $\mathbf{w}_1 = \mathbf{w}_2$

$\therefore \mathrm{W}_1 : \mathrm{w} = 1 : 1$

Q.18 Answer:

$16 : 1 : 2$

Solution:

According to Avogadro's law "At same temperature and pressure, Volume $\propto$ **no. of moles"** $\mathrm{n}_{\mathrm{H}_2} = \frac{\mathrm{w}}{2}; \quad \mathrm{n}_{\mathrm{O}_2} = \frac{\mathrm{w}}{32}; \mathrm{n}_{\mathrm{CH}_4} = \frac{\mathrm{w}}{16}$

$\because \mathrm{V}_{\mathrm{H}_2} : \mathrm{V}_{\mathrm{O}_2} : \mathrm{V}_{\mathrm{CH}_4} = \mathrm{n}_{\mathrm{H}_2} : \mathrm{n}_{\mathrm{O}_2} : \mathrm{n}_{\mathrm{CH}_4}$

$= \frac{\mathrm{w}}{2} : \frac{\mathrm{w}}{32} : \frac{\mathrm{w}}{16} = 16 : 1 : 2$

Q.19 Answer:

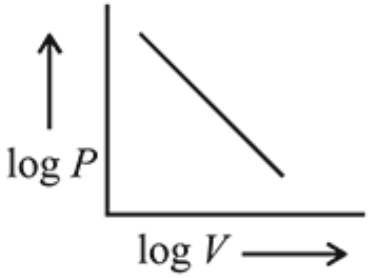
Straight line with a negative slope.

Solution:

According to Boyle's law, $PV =$ **constant** $\therefore \log P + \log V =$ **constant**

$\log P = -\log V +$ **constant**

Hence, the plot of $\log P$ **vs** $\log V$ **is straight line with negative slope.**



Q.20 Answer:

$\mathrm{CO}_2 = 300$ mL; $\mathrm{CO} = 400$ mL

Solution:

$\mathrm{CO}_2 + \mathrm{C} \longrightarrow 2\mathrm{CO}$

Stoichoimetry ratio is $1 : 2$ **AT STP,** $P = 1$ atm, $T = 273$ K, $R = 0.0821$

@IITJEE_Advanced

Initial moles of CO_2; $n\left(CO_2 \text{ initial }\right)=\frac{PV}{RT}$

$$=\frac{1 \times 0.5}{0.0821 \times 273}=0.022 \text{ mole}$$

In final mixture no. of moles; $n\left(CO_2 / CO \text{ mixture }\right)$

$$=\frac{1 \times 0.7}{0.0821 \times 273}=0.031$$

Increase in volume is $by = 0.031-0.022=0.009$ **mole of gas Final no. of moles of** CO **i.e.**

$n_{(\text{CO final})}$

$n_{(\text{CO final })}=2 n_{(\text{CO}_2 \text{ initial })}-n_{(\text{CO}_2 \text{ final })}$

$=2\left(0.022-\mathrm{n}_{(\text{CO}_2 \text{ final })}\right.$

$n_{(\text{CO final })}=0.044-2 n_{(\text{CO}_2 \text{ final}}$

$\therefore$ **Now,** $n_{(\text{CO final})}+n_{(\text{CO final})}=0.031 \, n_{(\text{CO}_2 \text{ final })}=0.031-n_{(\text{CO final })}$

Substituting (ii) in eq. (i) $n_{(\text{CO final })}=0.044-2\left[0.031-n_{(\text{CO final})}\right.$

$n_{(\text{CO final })}=0.044-0.062+2 n_{(\text{CO final })}$

$n_{(\text{CO final })}=0.018 \text{ mol.}$

Volume of $CO=V=\frac{nRT}{P}=\frac{0.018 \times 0.0821 \times 273}{1}$

$=0.40$ **Litre and volume of** $CO_2=0.7$ **litre** -0.4 **litre** $=0.3$ **litre** $\therefore CO_2=300 \text{ mL}, CO=400 \text{ mL}$

@IITJEE_Advanced